College Code- 240

ABACUS INSTITUTE OF ENGINEERING AND MANAGEMENT
(APPROVED BY AICTE AND AFFILIATED TO MAKAUT)

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NAME: SAHELI MONDAL

ROLL. NO.- 24000122001 BRANCH Computer Science and Engineering

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Group-B

2.a b. Bit rate: The number of bits transmitted per second is called bit rate (bps).

Baud rate: The number of signal changes per sec in communication channel called baud rate.

~~Bit rate~~

Bits in each signal unit = 4 bits

Signal units are sent per second = 1000 signal.

Bit rate = 1000×4 = 4000 bps

$$\text{Baud rate} = \frac{\text{Bit rate}}{\text{bits/signal unit}} = \frac{4000}{4} = 1000 \text{ bits}$$

2.e. 10 bit sequence 1010011110

divisor polynomial $= x^3+x^2+1$

∴ divisor = 1101

```
1101 ) 1010011110000
       1101
       ----------
       0111011110000
        1101
       ----------
       0011111100000
         1101
       ----------
       0010110000
         1101
       ----------
       0110000
        1101
       ----------
       000100
         1101
       ----------
        0101  CRC
```

```
1101 ) 1010011110101
       1101
       ----
       0111 0111101 01
        1101
        ----
        001111110101
          1101
          ----
          0010110101
           1101
           ----
           01100101
            1101
            ----
             1101
             1101
             ----
             0000
```

2.a ~~Different types of topologies~~ in computer network.

2.f. Stop and wait protocol

① The sender sends one frame and waits for an acknowledgement from the sender.

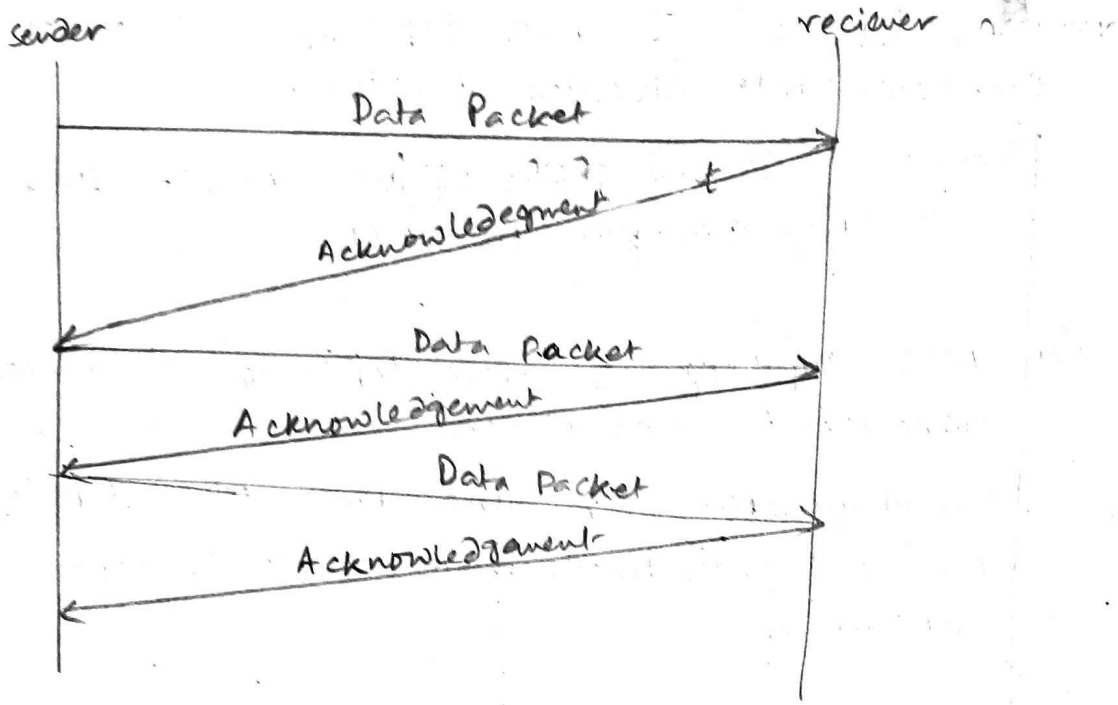
② the sender sends the next frame only after recieving the acknowledgment.

③ If the sender does not get the acknowledgment, it stops the process to send the next frame.

Its characteristics.

① Its efficiency is low, as it needs to wait for reciever's acknowledgment.

② Works on sample retransmission.



2.a. Different type of topologies:

① Bus topology: In this topology all devices are ~~conn~~ connected in one central hub called bus.

advantages: — ① easy to install and maintaince
(ii) flexible.

disadvantage: ① need more cables than star topology to connect device.
(ii) The central failure makes the full system data loss.

② Star topology: In this topology all devices are connected to one hub or switch. Data goes sender to switch then receiver.

advantages: ① when one ~~one~~ device affected others are unaffected.

disadvantages: ① If hub is affected it leads to system failure.

③ Ring topology: In ring topology all devices are connected into circular function.

Advantages: ① One device failure doesn't affect to others.

Disadvantages: ① Complex to install.

④ Mesh topology: In mesh topology all devices are connected to each other.

Advantages: ① More reliable than other topology.

Disadvantages: ① Needs more cable to connect to each other.

⑤ Tree topology: Tree topology is hybrid topology. Two topology (Bus + Star) combines and make tree topology. It has root and several branches make tree. like hierarchy.

Advantage: Highly secure

Disadvantage: Cost effective.

Group-A

1.a. ii $n(n-1)/2$

b. ii Medium

c. iii security

d. iii infinite

e. i Simplex

f. ii cyclic redundancy check

g.